

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3331

3 Credits: 3

Engineering Science

Source-grounded

Analog & Digital Electronics

□ To introduce various semiconductor diodes and application circuits.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3331 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3331.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	3331
Course title	Analog & Digital Electronics
Semester	3 Credits: 3
Course category	Engineering Science
Periods per week	3 (L:3, T:0, P:0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- To introduce various semiconductor diodes and application circuits.
- To illustrate the working of different transistors.
- To introduce various linear and non linear wave shaping circuits.

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding Junction Transistor and transistor amplifier. Summarize the working principles of JFET,	See official syllabus
CO2	11 Understanding MOSFET and UJT. Design and implement Combinational and	See official syllabus
CO3	Sequential logic circuits. 11 Applying Describe the process of Analog to Digital conversion	See official syllabus
CO4	and Digital to Analog conversion. 11 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	transistor amplifier.	5	As specified
Module 2	Summarize the working principle of JFET, MOSFET and UJT	4	As specified
Module 3	Design and implement combinational and sequential circuits.	5	As specified
Module 4	Describe the process of analog to digital and digital analog conversion	3	As specified

MODULE 1

transistor amplifier.

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: transistor amplifier.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Describe the physical structure of BJT

Describe the physical structure of BJT is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the physical structure of BJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the physical structure of bjt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the physical structure of BJT means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
 Define the physical structure of BJT
 definition/meaning.
 steps, application
 Technical terms
 English-
 .

▼ Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations

Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the working principle of bjt. 2 understanding summarize various transistor configurations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations definition/meaning . steps, application . Technical terms English- .

▼ 2 Understanding and gain relations. Illustrate the characteristics ofBJT

2 Understanding and gain relations. Illustrate the characteristics ofBJT is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and gain relations. Illustrate the characteristics ofBJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and gain relations. illustrate the characteristics ofbjt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding and gain relations. Illustrate the characteristics ofBJT means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding and gain relations. Illustrate the characteristics ofBJT definition/meaning. steps, application. Technical terms English- .

▼ 3 Understanding Explain transistor action and amplification

3 Understanding Explain transistor action and amplification is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Explain transistor action and amplification using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding explain transistor action and amplification should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding Explain transistor action and amplification means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding Explain transistor action and amplification ഏകദേശം definition/meaning ഉൾപ്പെടെ. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

▼ 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain β_{DC} and β_{AC} - relationship between β_{DC} and β_{AC} - effect of temperature in leakage current Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CE configuration Transistor action - Transistor as amplifier.

1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain β_{DC} and β_{AC} - relationship between β_{DC} and β_{AC} - effect of temperature in leakage current Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CE configuration Transistor action - Transistor as amplifier. is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain β_{DC} and β_{AC} -

relationship between β and β_{DC} - effect of temperature in leakage current Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CE configuration Transistor action - Transistor as amplifier. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding bipolar junction transistors - structure - npn and pnp - symbol - unbiased transistor transistor biasing - modes of operation - active, saturation and cut off - operation of npn transistor - transistor currents transistor configurations - current gain β_{DC} and β - relationship between β and β_{DC} - effect of temperature in leakage current input and output characteristics of common base and common emitter configuration - cut off, active and saturation regions - comparison of three configurations - importance of CE configuration transistor action - transistor as amplifier. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain β_{DC} and β - relationship between β and β_{DC} - effect of temperature in leakage current Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CE configuration Transistor action - Transistor as amplifier. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain β_{DC} and β_{AC} - relationship between β_{DC} and β_{AC} - effect of temperature in leakage current Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CE configuration Transistor action - Transistor as amplifier. β_{DC} definition/meaning β_{AC} . β_{DC} β_{AC} β_{DC} , steps, application β_{DC} β_{AC} β_{DC} . Technical terms English- Malayalam β_{DC} .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize the working principle of JFET, MOSFET and UJT

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the working principle of JFET, MOSFET and UJT
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding MOSFET and UJT Explain the working principle of JFET,

3 Understanding MOSFET and UJT Explain the working principle of JFET, is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding MOSFET and UJT Explain the working principle of JFET, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mosfet and ujt explain the working principle of jfet, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding MOSFET and UJT Explain the working principle of JFET, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding MOSFET and UJT Explain the working principle of JFET, definition/meaning, steps, application. Technical terms English- Malayalam.

▼ 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET,

3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mosfet and ujt illustrate the characteristics of jfet, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 4 Understanding MOSFET and UJT

4 Understanding MOSFET and UJT is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding MOSFET and UJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding mosfet and ujt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding MOSFET and UJT means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding MOSFET and UJT
 പരിശീലനം definition/meaning. പരിശീലനം പരിശീലനം
 പരിശീലനം, steps, application പരിശീലനം പരിശീലനം പരിശീലനം. Technical terms
 English- പരിശീലനം പരിശീലനം പരിശീലനം.

▼ Compare BJT and FET 1 Remembering Junction Field Effect

Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator

Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare bjt and fet 1 remembering junction field effect transistors: structure - symbol - operation of n channel jfet - drain characteristics - transfer characteristics - comparison with bjt. mosfet: structure - symbol - principle of operation (n channel depletion type only), drain characteristics - transfer characteristics. ujt: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - vi characteristics. ujt relaxation oscillator should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator ഓസിലേറ്റർ ഓസിലേറ്റർ definition/meaning ഓസിലേറ്റർ. ഓസിലേറ്റർ ഓസിലേറ്റർ, steps, application ഓസിലേറ്റർ ഓസിലേറ്റർ. Technical terms English- Malayalam ഓസിലേറ്റർ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Design and implement combinational and sequential circuits.

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design and implement combinational and sequential circuits.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,

Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, simplification of logic function using k-map 1 remembering explain the working of multiplexer, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, □□□□□□□□□□□□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□□□.

▼ 2 Understanding demultiplexer, adder and subtractor.

2 Understanding demultiplexer, adder and subtractor. is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding demultiplexer, adder and subtractor. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding demultiplexer, adder and subtractor. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding demultiplexer, adder and subtractor. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding demultiplexer, adder and subtractor. ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English- ഫലപ്രസാദം ഫലപ്രസാദം.

▼ Develop BCD adder and Carry look head 2 Applying adder

Develop BCD adder and Carry look head 2 Applying adder is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop BCD adder and Carry look head 2 Applying adder using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop bcd adder and carry look head 2 applying adder should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop BCD adder and Carry look head 2 Applying adder means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
2 Applying adder
 definition/meaning
 , steps, application
 Technical terms English-

▼ Understand the working of different Latches 3 Applying and Flipflops

Understand the working of different Latches 3 Applying and Flipflops is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the working of different Latches 3 Applying and Flipflops using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the working of different latches 3 applying and flipflops should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Understand the working of different Latches 3 Applying and Flipflops means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Understand the working of different Latches 3 Applying and Flipflops definition/meaning . steps, application . Technical terms English- .

▼ **Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops, Shift registers, Synchronous and asynchronous counters. Sequence generator.**

Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops, Shift registers, Synchronous and asynchronous counters. Sequence generator. is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops, Shift registers, Synchronous and asynchronous counters. Sequence generator. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design shift registers, synchronous and 3 applying asynchronous counters using flipflops simplification of logic functions using k-map, minimization of logical functions. don't care conditions, multiplexer, de-multiplexer/decoders, adders, subtractors, bcd arithmetic, carry look ahead adder. 1 bit memory, clocked sr jk d flipflops, shift registers, synchronous and asynchronous counters. sequence generator. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops, Shift registers, Synchronous and asynchronous counters. Sequence generator. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops, Shift registers, Synchronous and asynchronous counters. Sequence generator. definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Describe the process of analog to digital and digital analog conversion

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the process of analog to digital and digital analog conversion
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding conversion. Explain the working of various digital

3 Understanding conversion. Explain the working of various digital is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding conversion. Explain the working of various digital using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding conversion. explain the working of various digital should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 3 Understanding conversion. Explain the working of various digital means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3 Understanding conversion. Explain the working of various digital conversion definition/meaning conversion. conversion conversion conversion, steps, application conversion conversion conversion. Technical terms English- conversion conversion.

▼ 4 Understanding to analog convertors Explain the working of various analog

4 Understanding to analog convertors Explain the working of various analog is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding to analog convertors Explain the working of various analog using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding to analog convertors explain the working of various analog should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 4 Understanding to analog convertors Explain the working of various analog means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding to analog convertors Explain the working of various analog definition/meaning , steps, application Technical terms English-

▼ **4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 <https://www.electronics-tutorials.ws> 2 <https://www.elprocus.com> 3 <http://www.brainkart.com/menu/electronics-engineering/> 4 <https://www.electrical4u.com> 5 <https://www.tutorialspoint.com>**

4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 <https://www.electronics-tutorials.ws> 2 <https://www.elprocus.com> 3 <http://www.brainkart.com/menu/electronics-engineering/> 4 <https://www.electrical4u.com> 5 <https://www.tutorialspoint.com> is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 <https://www.electronics-tutorials.ws> 2 <https://www.elprocus.com> 3 <http://www.brainkart.com/menu/electronics-engineering/> 4 <https://www.electrical4u.com> 5 <https://www.tutorialspoint.com> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding to digital convertors analog to digital conversion: sampling, quantization, encoding digital to analog convertors: weighted resistor, r-2r ladder analog to digital convertors: counter type, successive approximation, dual slope, example of a to d and d to a convertor ics text /reference: t/r booktitle/author t1 r s sedha - a text book of applied electronics - s chand n n bhargava, kulshreshtha and s c gupta - basic electronics and linear r1 circuits- tmh r2 robert boylestad - electronic devices and circuits - phi anand kumar- fundamentals of digital circuits r3 r4 david a bell - electronic devices and circuits - phi online resources: sl. no website link 1 <https://www.electronics-tutorials.ws> 2 <https://www.elprocus.com> 3 <http://www.brainkart.com/menu/electronics-engineering/> 4

<https://www.electrical4u.com> 5 [https://www. tutorialspoint.com](https://www.tutorialspoint.com) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of

A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 <https://www.electronicstutorials.ws> 2 <https://www.elprocus.com> 3

<http://www.brainkart.com/menu/electronics-engineering/> 4

<https://www.electrical4u.com> 5 <https://www.tutorialspoint.com>

പ്രാഥമികമായി പദപരിചയം definition/meaning നൽകുന്നു. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം പദപരിചയം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Describe the physical structure of BJT. (3 marks)

► Define or explain Explain the working principle of BJT. 2
Understanding Summarize various transistor configurations. (3 marks)

► Define or explain 2 Understanding and gain relations. Illustrate the characteristics of BJT. (3 marks)

► Define or explain 3 Understanding Explain transistor action and amplification. (3 marks)

Module 2

► Define or explain 3 Understanding MOSFET and UJT Explain the working principle of JFET,. (3 marks)

► Define or explain 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET,. (3 marks)

► Define or explain 4 Understanding MOSFET and UJT. (3 marks)

► Define or explain Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT.

MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator. (3 marks)

Module 3

► Define or explain Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,. (3 marks)

► Define or explain 2 Understanding demultiplexer, adder and subtractor.. (3 marks)

► Define or explain Develop BCD adder and Carry look head 2 Applying adder. (3 marks)

► Define or explain Understand the working of different Latches 3 Applying and Flipflops. (3 marks)

Module 4

► Define or explain 3 Understanding conversion. Explain the working of various digital. (3 marks)

► Define or explain 4 Understanding to analog convertors Explain the working of various analog. (3 marks)

► Define or explain 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits-

TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 <https://www.electronics-tutorials.ws> 2 <https://www.elprocus.com> 3 <http://www.brainkart.com/menu/electronics-engineering/> 4 <https://www.electrical4u.com> 5 <https://www.tutorialspoint.com>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Describe the physical structure of BJT.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding MOSFET and UJT Explain the working principle of JFET,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,.**

Module 4

1-mark definition: State the meaning of **3 Understanding conversion. Explain the working of various digital.**

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link <https://www.electronics-tutorials.ws>
<https://www.elprocus.com>
- <http://www.brainkart.com/menu/electronics-engineering/>
<https://www.electrical4u.com> <https://www.tutorialspoint.com>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTR syllabus PDF for Course 3331. Verify institutional updates against the current official curriculum.